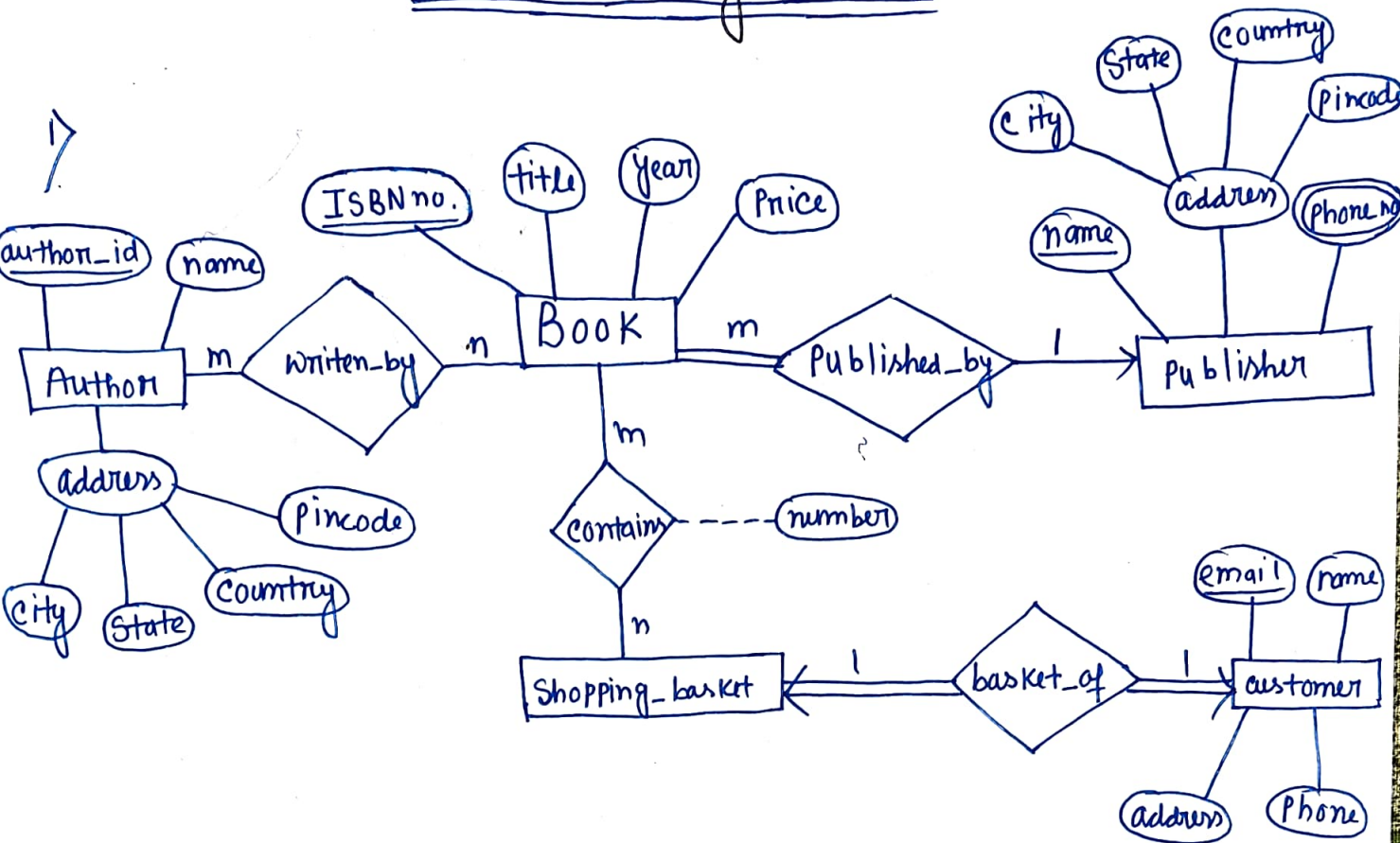


ITD Assignment -1



2) Relationship schema—

customer (cid, cname, city, state, pincode, DOB)

Loan (lno, lamount, brid FK)

Payment (pno, lno, pdate, Pamount)

Account (accno, balance, brid FK)

Branch (brid, brname, city, assets)

Saving account (accno, Interest-rate)

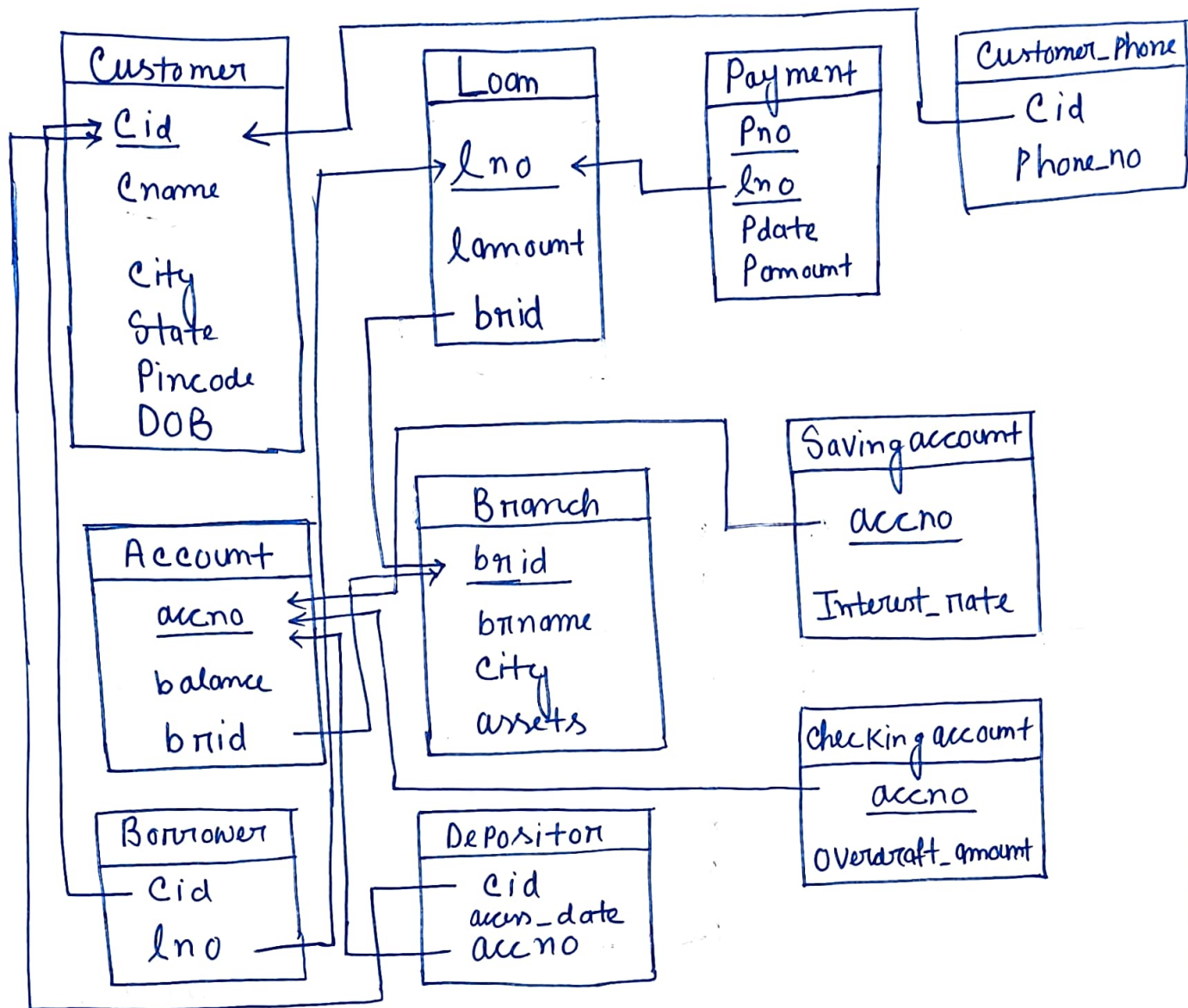
Checking account (accno, Overdraft-amount)

Borrower (cid FK, lno FK)

Depositor (cid FK, accno FK, access-date)

customer-Phone (cid FK, Phone)

3)



4) $R(A, B, C, D, E, F)$

$A \rightarrow BC$

$CD \rightarrow E$

$B \rightarrow D$

$E \rightarrow A$

}

∴ candidate keys = {AF, EF}

closure

$A^+ = ABCDE$

$B^+ = BD$

$C^+ = C$

$D^+ = D$

$E^+ = EABCD$

$F^+ = F$

$(AF)^+ = ABCDEF$

$(EF)^+ = EFABCD$

$(BF)^+ = BDF$

$(CF)^+ = CF$

$(DF)^+ = DF$

5) Student mark (regd, name, course_id, title, grade)

$F(\text{regd}, \text{name}, \text{course_id}, \text{title}, \text{grade})\{$

$\text{regd} \rightarrow \text{name}$

$\text{course_id} \rightarrow \text{title}$

$\{ (\text{regd}, \text{course_id}) \rightarrow \text{grade}$

$(\text{regd}, \text{course_id})^+ = \{ \text{regd}, \text{course_id}, \text{name}, \text{title}, \text{grade} \}$

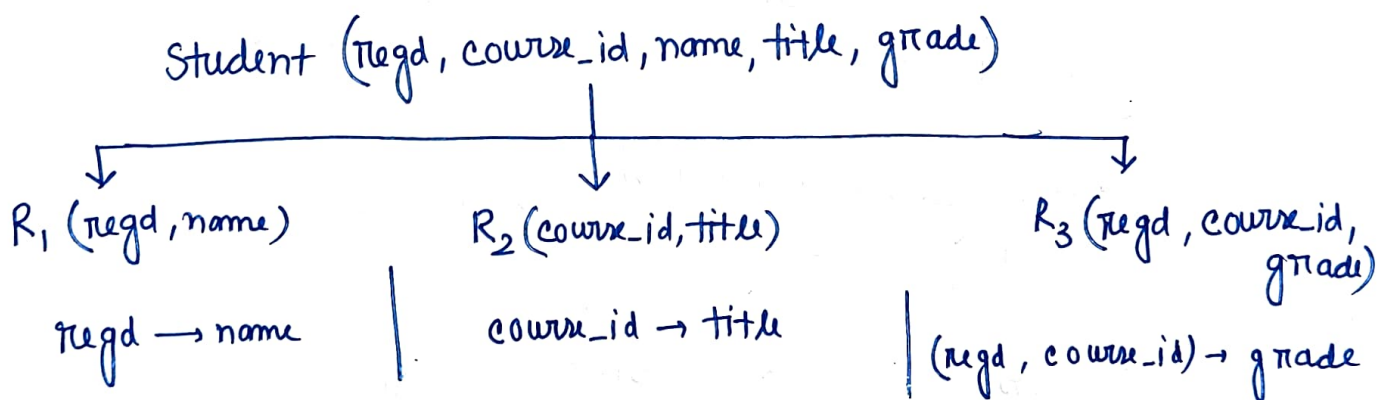
$P = \{ \text{regd}, \text{course_id} \}$ $CK = \{ (\text{regd}, \text{course_id}) \}$

$NP = \{ \text{grade}, \text{name}, \text{title} \}$

$\therefore$ All attributes are atomic $\Rightarrow$ 1NF

$\left. \begin{array}{l} \text{regd} \rightarrow \text{name} \\ \text{course_id} \rightarrow \text{title} \end{array} \right\} \text{Partial Dependency}$

$\therefore F$ is not in 2NF



$\therefore R_1 \cup R_2 \cup R_3 = \text{Student}$

$R_1 \cap R_3 = \{ \text{regd} \} \rightarrow \text{superkey of } R_1$

$R_2 \cap R_3 = \{ \text{course_id} \} \rightarrow \text{superkey of } R_2$

$\therefore R_1, R_2, R_3$ are lossless

$\therefore$ All FD's Preserved

$\therefore$ This is 2NF Ans

6) Book (Title, Author, Catalog-no, Publisher, Year, Price)

$F \{ (Title, Author) \rightarrow (Catalog-no, Price),$

$Catalog-no \rightarrow Title$

$Catalog-no \rightarrow Publisher$

$\{ Catalog-no \rightarrow Year$

$(Title, Author)^+ = \{ Title, Author, Catalog-no, Price, Publisher, Year \}$

$(Catalog-no, Author)^+ = \{ Catalog-no, Author, Title, Publisher, Year, Price \}$

$\therefore CK = \{ (Title, Author), (Catalog-no, Author) \}$

$P = \{ Title, Author, Catalog-no \}$

$NP = \{ Publisher, Year, Price \}$

$\therefore$ A relation is in 3NF if for $X \rightarrow Y$, X is a superkey on Y is prime

$\therefore \left. \begin{array}{l} Catalog-no \rightarrow Publisher \\ Catalog-no \rightarrow Year \end{array} \right\} \text{Violates the 3NF condition}$

$\therefore$ This is not in 3NF

Book (Title, Author, Catalog-no, Publisher, Year, Price)

$R_1 (Catalog-no, Title, Publisher, Year)$

$Catalog-no \rightarrow Title$
 $Catalog-no \rightarrow Publisher$
 $Catalog-no \rightarrow Year$

$R_2 (Author, Price, Title, Catalog-no)$

$(Title, Author) \rightarrow (Catalog-no, Price)$

$$\therefore CK = \{ \text{Catalog_no} \}$$

$\therefore \text{Catalog_no}$ is superkey of R_1

$$\therefore CK = \{ (\text{Title}, \text{Author}) \}$$

$(\text{Title}, \text{Author})$ is superkey of R_2

$\therefore$ It satisfies the 3NF condition

$$\therefore R_1 \cup R_2 = \text{Book}$$

$$R_1 \cap R_2 = (\text{Catalog_no}, \text{Title}) \rightarrow \text{superkey of } R_1$$

$\therefore$ Lossless decomposition

$\therefore R_1$ & R_2 is in 3NF Ans

$$\Rightarrow \pi(A, B, C, D, E, G): F = \begin{cases} A \rightarrow BCD, \\ BC \rightarrow DE, \\ B \rightarrow D, \\ D \rightarrow A \end{cases}$$

(a) To prove: AG is a super key

(i) $A \rightarrow BCD$ (given)

(ii) $A \rightarrow B$ (Decomposition)

(iii) $A \rightarrow C$ (Decomposition)

(iv) $A \rightarrow D$ (Decomposition)

(v) $A \rightarrow BC$ (union of (ii) & (iii))

(vi) $BC \rightarrow DE$ (given)

(vii) $A \rightarrow DE$ (transitivity (v) & (vi))

(viii) $A \rightarrow E$ (Decomposition)

$$(ix) A \rightarrow BCDE \text{ (union of (i) \& (viii))}$$

$$(x) AG \rightarrow A, AG \rightarrow G \text{ (reflexivity)}$$

$$(xi) AG \rightarrow BCDEG \text{ (augment G in (ix))}$$

$$(xii) AG \rightarrow ABCDEG \text{ (union of (ix) \& (xi))}$$

$\therefore AG$ is a super key (Proved)

$$(b) F \{ \begin{array}{l} A \rightarrow BCD, \\ BC \rightarrow DE \\ B \rightarrow D \\ D \rightarrow A \end{array}$$

$$(i) \begin{array}{l} A \rightarrow B \\ A \rightarrow C \\ A \rightarrow D \\ BC \rightarrow D \\ BC \rightarrow E \\ B \rightarrow D \\ D \rightarrow A \end{array}$$

$$(ii) \begin{array}{l} A \rightarrow B \\ A \rightarrow C \\ BC \rightarrow D \\ BC \rightarrow E \\ B \rightarrow D \\ D \rightarrow A \end{array}$$

Transitivity removed

$$(iii) \begin{array}{l} A \rightarrow B \\ A \rightarrow C \\ BC \rightarrow E \\ B \rightarrow D \\ D \rightarrow A \end{array}$$

$\therefore$ extraneous $\rightarrow D$ removed

$$(iv) BC \rightarrow E$$

$$\begin{array}{l} \downarrow \\ B^+ = BDACE \\ C^+ = C \end{array}$$

$\therefore$ can remove C

$$\therefore B \rightarrow DE$$

$\therefore$ final FD's. —

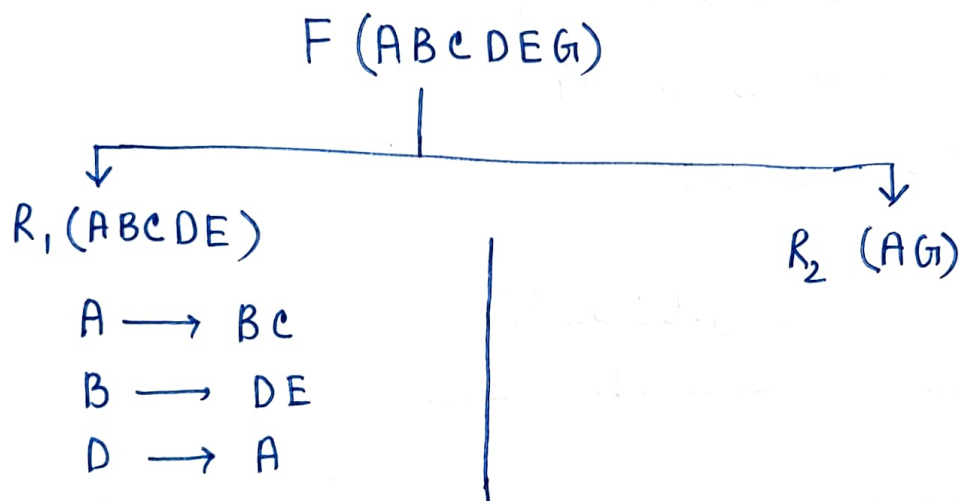
$$\left. \begin{array}{l} A \rightarrow B \\ A \rightarrow C \\ B \rightarrow E \\ B \rightarrow D \\ D \rightarrow A \end{array} \right\} \begin{array}{l} A \rightarrow BC \\ B \rightarrow DE \\ D \rightarrow A \end{array}$$

Ans

(c) $F \{ A \rightarrow BC, \\ (ABCEG) \quad B \rightarrow DE, \\ \quad \quad \quad D \rightarrow A$

$$\begin{array}{l|l} A^+ = ABCDE & (AG)^+ = AGBCDE \\ B^+ = BDEAC & (BG)^+ = BGDEAC \\ D^+ = DABCE & (DG)^+ = DGABCE \end{array}$$

$$\begin{aligned} CK &= \{AG, BG, DG\} \\ P &= \{A, B, D, G\} \\ NP &= \{C, E\} \end{aligned}$$



$$\therefore R_1 \cup R_2 = F$$

$$R_1 \cap R_2 = A \rightarrow \text{superkey of } R_1$$

$$\therefore \left. \begin{array}{l} R_1(ABCDE) \\ R_2(AG) \end{array} \right\} \text{ This is now in 3NF form}$$

(d) $\pi(ABCDEG) \{$

$A \rightarrow BCD,$

$BC \rightarrow DE,$

$B \rightarrow D,$

$D \rightarrow A$

$\}$

$A^+ = ABCDE$

$B^+ = BDACE$

$D^+ = DABCE$

$(AG)^+ = AGBCDE$

$(BG)^+ = BGDAEC$

$(DG)^+ = DGABCE$

$CK = \{AG, BG, DG\}$

$P = \{A, B, D, G\}$

$NP = \{C, E\}$

$\pi(ABCDEG)$

$R_1(ABCDE)$

$A \rightarrow BCD$

$BC \rightarrow DE$

$B \rightarrow D$

$D \rightarrow A$

$CK = \{A, B, D\}$

$\therefore R_1$ in BCNF

$R_2(AG)$

R_2 in BCNF

$\therefore R_1 \cup R_2 = \pi$

$R_1 \cap R_2 = A \rightarrow$ Superkey of R_1

$\therefore$ lossless decomposition

8) (a)

$\pi(PAN, PI, DI, DRUG, QTY, COST)\{$

$PAN \rightarrow PI,$

$PI \rightarrow DI,$

$(PI, DRUG) \rightarrow QTY,$

$\{ (DRUG, QTY) \rightarrow COST$

$$PAN^+ = \{PAN, PI, DI\} \quad (PI, DRUG)^+ = \{PI, DRUG, QTY, COST, DI\}$$

$$PI^+ = \{PI, DI\}$$

$$(DRUG, QTY)^+ = \{DRUG, QTY, COST\}$$

$$(PAN, PI, DRUG)^+ = \{PAN, PI, DRUG, DI, QTY, COST\}$$

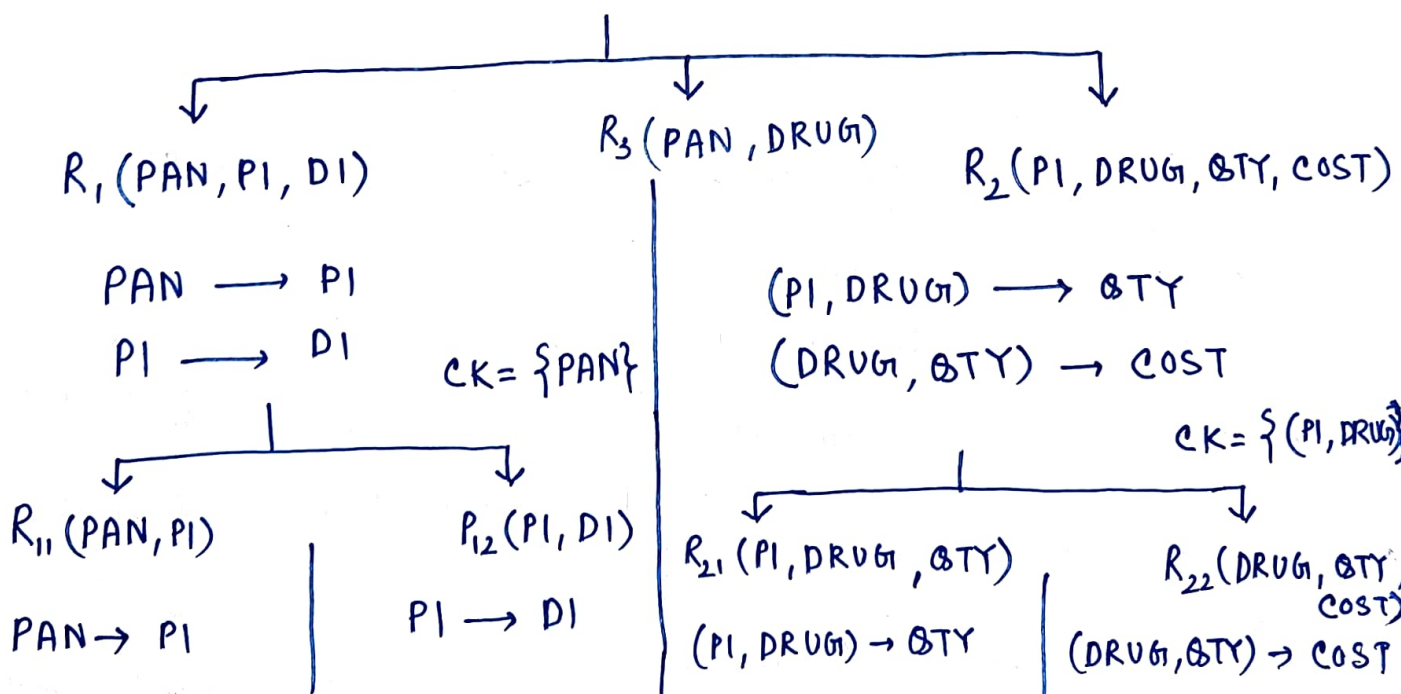
$$(PAN, DRUG)^+ = \{PAN, DRUG, PI, DI, QTY, COST\}$$

$$\therefore CK = \{ (PAN, DRUG) \}$$

$$P = \{PAN, DRUG\}$$

$$NP = \{PI, DI, QTY, COST\}$$

$\pi(PAN, PI, DI, DRUG, QTY, COST)$



$R_{11} (PAN, PI)$
 $R_{12} (PI, DI)$
 $R_{21} (PI, DRUG, QTY)$
 $R_{22} (DRUG, QTY, COST)$
 $R_3 (PAN, DRUG)$

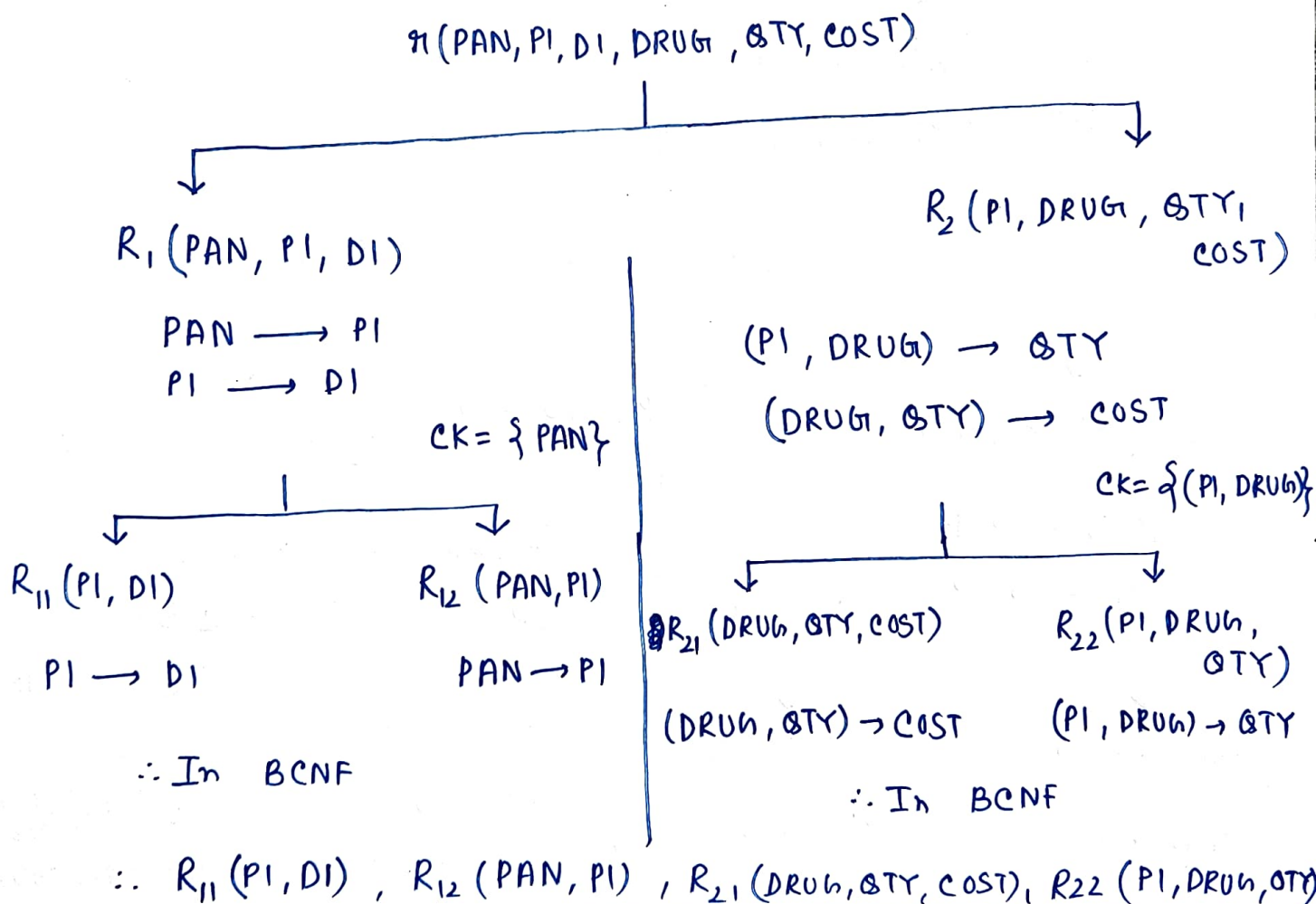
This the 3NF decomposition

(b) $\pi(PAN, PI, DI, DRUG, QTY, COST)\}$

$PAN \rightarrow PI,$
 $PI \rightarrow DI,$
 $(PI, DRUG) \rightarrow QTY,$
 $(DRUG, QTY) \rightarrow COST$

}

$CK = \{ (PAN, DRUG) \}$



9) $R(\text{VOLUME, NUMBER, STARTPAGE, ENDPAGE, TITLE, YEAR, PRICE})$

$(\text{VOLUME, NUMBER, STARTPAGE, ENDPAGE}) \rightarrow \text{TITLE}$

$(\text{VOLUME, NUMBER}) \rightarrow \text{YEAR}$

$\{ (\text{VOLUME, NUMBER, STARTPAGE, ENDPAGE}) \rightarrow \text{PRICE} \}$

$CK = \{ (\text{VOLUME, NUMBER, STARTPAGE, ENDPAGE}) \}$

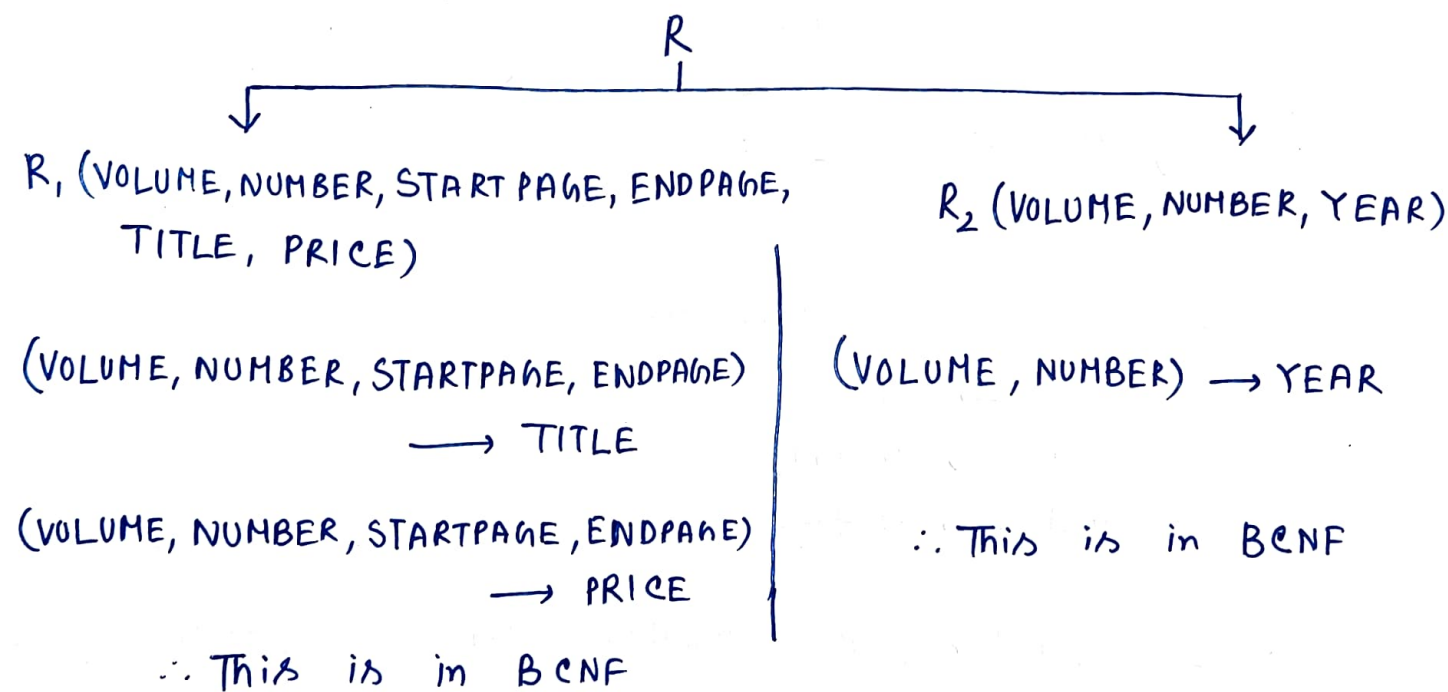
$P = \{ \text{VOLUME, NUMBER, STARTPAGE, ENDPAGE} \}$

$NP = \{ \text{TITLE, YEAR, PRICE} \}$

$\therefore (\text{VOLUME, NUMBER}) \rightarrow \text{YEAR}$ (Partial Dependency exist)

$\therefore$ This is not in 2NF

$\therefore$ The old schema is in 1NF



$\therefore$ Highest NF = BCNF
Weakest NF = 2NF } Ans

10/ (a) books (accessionno, isbn, title, author, publisher) {
 accessionno $\rightarrow$ isbn,
 isbn $\rightarrow$ title,
 isbn $\rightarrow$ publisher,
 { isbn $\rightarrow$ author,

CK = { accessionno }

P = { accessionno }

NP = { isbn, title, publisher, author }

isbn $\rightarrow$ title
 isbn $\rightarrow$ publisher
 isbn $\rightarrow$ author } doesn't follow 3NF condition

$\therefore$ Not in 3NF

$\therefore$ There is no partial dependency

$\therefore$ Highest NF = 2NF

Users (userid, name, deptid, deptname) {

userid $\rightarrow$ name,
 userid $\rightarrow$ deptid,
 { deptid $\rightarrow$ deptname

CK = { userid } P = { userid } NP = { name, deptid, deptname }

$\therefore$ ~~userid~~ deptid } doesn't follow 3NF condition
 deptid $\rightarrow$ deptname

$\therefore$ There is no partial dependency

$\therefore$ Highest NF = 2NF

Ans

(b)

books (accessionno, isbn, title, author, Publisher)

R_1 (isbn, title, Publisher, author)

isbn $\rightarrow$ title

isbn $\rightarrow$ Publisher

isbn $\rightarrow$ author

CK = {isbn}

$\therefore$ This is in BCNF

R_2 (accessionno, isbn)

accessionno $\rightarrow$ isbn

$\therefore$ This is in 4NF

R_1 (isbn, title, Publisher, author)

R_{11} (isbn, author)

isbn $\rightarrow$ author

$\therefore$ This is in 4NF

R_{12} (isbn, title, Publisher)

isbn $\rightarrow$ title

isbn $\rightarrow$ Publisher

$\therefore$ This is in 4NF

$\therefore R_{11} \cup R_{12} \cup R_2 = \text{books}$

$R_{11} \cap R_{12} = \text{isbn} \rightarrow \text{PK of } R_1$

$R_1 \cap R_2 = \text{isbn} \rightarrow \text{PK of } R_1$

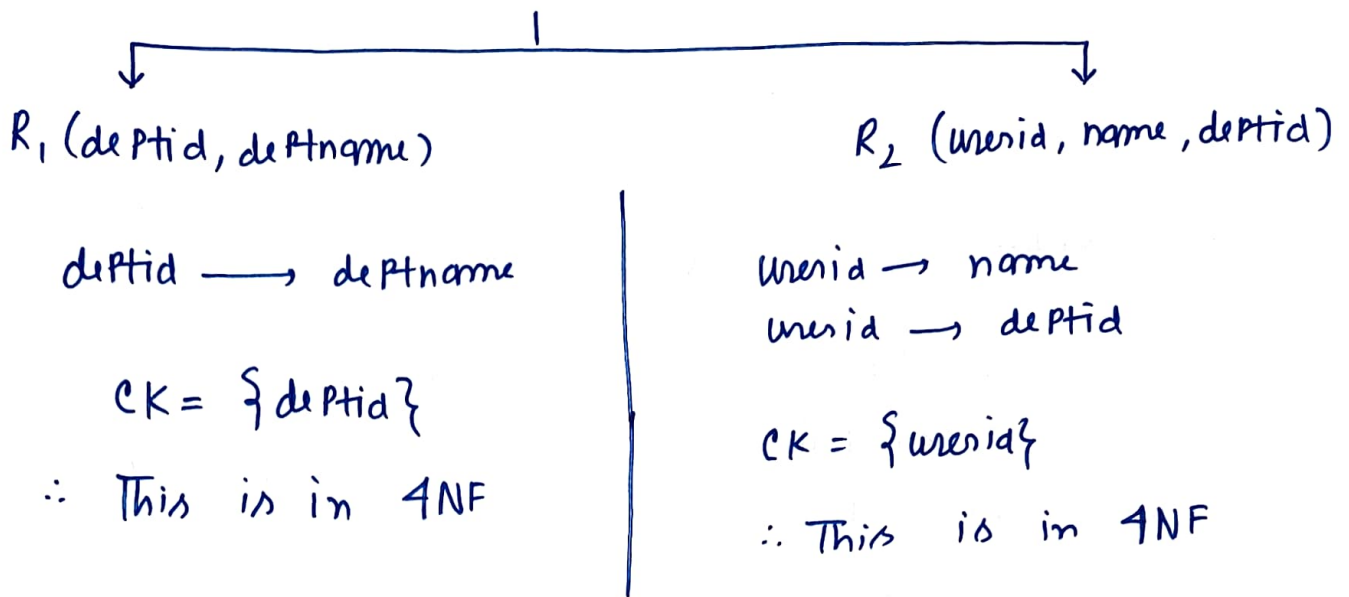
$\therefore R_{11}$ (isbn, author)

R_{12} (isbn, title, Publisher)

R_2 (accessionno, isbn)

} 4NF decomposition

users (userid, name, deptid, deptname)



$$\therefore R_1 \cup R_2 = users$$

$$R_1 \cap R_2 = deptid \rightarrow PK \text{ of } R_1$$

$$\therefore \left. \begin{array}{l} R_1 (deptid, deptname) \\ R_2 (userid, name, deptid) \end{array} \right\} \text{ 4NF decomposition}$$